

ME5 | Rubric | Spring 2026

Total: 100 points

ACC — 20 pts

Q1. Model Selection with Controls — 20 pts

Spec: Pick the right model and interpret the coefficient with "holding constant."

a) Which regression model? — 9 pts

Correct: C) $\text{employment} = \beta_0 + \beta_1 \cdot \text{min_wage} + \beta_2 \cdot \text{cost_of_living} + \varepsilon$

Answer	Pts	Reasoning
C	9	Correct
A	3	Simple regression. Knows min_wage is the predictor but ignores the control
D	5	Interaction model. Has both variables but misreads "holding constant" as "differs by"
B	0	Wrong predictor entirely

b) Interpret the coefficient on min_wage — 11 pts

Spec: One-unit change, in context, with "holding constant."

Correct: The change in employment associated with each additional unit of minimum wage, comparing two states with equal cost of living.

Element	Pts
Mentions "change in employment" (direction + magnitude of change in y)	3
Mentions "each additional unit of minimum wage" (one-unit change in x)	3
Mentions "comparing states with equal cost of living" (holding constant)	3
Anything written (baseline)	2

Bright line: The "holding constant" element is what separates Part 5 from Part 4. A student who writes a correct Part 4 interpretation (no "holding constant") gets 8/11. The 3-point gap tests whether they understand what multiple regression does.

Q2. Binary Predictor with Control — 20 pts

Spec: Write the model, interpret the binary coefficient with "holding constant," and explain why we need the control.

Correct model: $\text{uninsured_rate} = \beta_0 + \beta_1 \cdot \text{expansion} + \beta_2 \cdot \text{poverty_rate} + \varepsilon$

a) Write the regression model — 8 pts

Answer	Pts	Reasoning
$\text{uninsured_rate} = \beta_0 + \beta_1 \cdot \text{expansion} + \beta_2 \cdot \text{poverty_rate} + \varepsilon$	8	Correct
$y = \beta_0 + \beta_1 \cdot x_1 + \beta_2 \cdot x_2 + \varepsilon$	5	Structure correct, no context
Flipped outcome	4	Right structure, wrong variable on the left
Missing one predictor	4	Partial model
Missing error term but otherwise correct	7	Minor omission
Includes interaction ($\text{expansion} \times \text{poverty_rate}$)	5	Overcomplicated
Blank	0	

b) Interpret the coefficient on expansion — 6 pts

Spec: β_1 is the difference between expansion and non-expansion counties, holding poverty constant.

Correct: The difference in average uninsured rate between expansion and non-expansion counties, comparing two counties with equal poverty rates.

Element	Pts
Mentions "difference in average uninsured rate"	2
Mentions "expansion and non-expansion counties"	2
Mentions "comparing counties with equal poverty rate" (holding constant)	2

c) Why control for poverty rate? — 6 pts

Spec: Poverty is correlated with both expansion and uninsured rate, so it confounds.

Correct: Poorer counties may have higher uninsured rates AND be less likely to adopt expansion (or vice versa).

Answer	Pts	Reasoning
Identifies that poverty is related to both variables	6	Correct
Something generally in the right direction	3	Partial
Blank	0	

Q3. Reading Regression Output (Used Cars / Mileage + Accident) — 20 pts

Spec: Make predictions from the output and interpret the binary coefficient.

Grader note: Mileage is in thousands of miles, so 60,000 miles = 60. Many students will plug in 60,000 and get a huge negative number. Units error, not conceptual. See partial credit below.

a) Predicted price: 60k miles, no accident — 6 pts

Correct: $\text{PRICE} = 22000 + (-120) \times 60 + (-3200) \times 0 = 22000 - 7200 = \$14,800$

Answer	Pts	Reasoning
\$14,800 with work shown	6	Correct
\$14,800 without work	5	Correct but no verification
Arithmetic error but correct setup (all terms present)	4	Right method, execution error
Uses 60,000 instead of 60 (wrong units)	3	Correct setup, didn't read "thousands of miles"
Missing a term	2	
Blank	0	

b) Predicted price: 60k miles, with accident — 6 pts

Correct: $\text{PRICE} = 22000 + (-120)(60) + (-3200)(1) = 22000 - 7200 - 3200 = 11,600$. *Or equivalently* : $14,800 - 3,200 = 11,600$ (building on part a).

Answer	Pts	Reasoning
\$11,600 with correct setup	6	Correct
Correct setup but with math error	5	
Incorrect setup	3	
Blank	0	

c) Interpret coefficient on accident (-3200) — 8 pts

Spec: -3200 is the price difference between accident and no-accident cars, holding mileage constant.

Correct: The extra average cost of a car with an accident history (compared to one without), holding mileage constant.

Element	Pts
Mentions "extra (average) cost" or "lower price" (direction)	3
Mentions "accident history" (context)	3
Anything written (baseline)	2

Q4. Interaction Model (Study Hours × Tutoring / Exam Scores) — 20 pts

Spec: Compute the effect for each group and interpret the interaction coefficient.

Coefficients: $\beta_0 = 65$, $\beta_1 = 2.5$, $\beta_2 = 4$, $\beta_3 = 1.5$

a) Effect for non-tutored students — 6 pts

Correct: $\beta_1 = 2.5$

Answer	Pts	Reasoning
β_1 or 2.5	6	Correct
Incorrect	2	
Blank	0	

b) Effect for tutored students — 6 pts

Correct: $\beta_1 + \beta_3 = 2.5 + 1.5 = 4.0$

Answer	Pts	Reasoning
$\beta_1 + \beta_3$ or 4.0	6	Correct
β_3 or 1.5 (interaction only)	3	Thinks interaction IS the full effect
Anything else	2	
Blank	0	

c) Interaction coefficient MC (circle one) — 8 pts

Correct: B) An additional hour of studying increases exam scores by 1.5 more for tutored students than non-tutored students

Answer	Pts	Reasoning
B	8	Correct
A	3	Confuses interaction (β_3) with fixed effect (β_2)
C	2	Reads 1.5 as a multiplier, not an additive difference
D	2	Confuses interaction with full effect for the tutored group